Website Build Prompt

Research laboratory website: Zoonotic Diseases Group, ICAR-NIHSAD, Bhopal

Principal Investigator: Dr. Ashwin Ashok Raut | Prepared October 2026, revised with the full CV (April 2026)

How to use this document. Give Part A (the instructions) and Part B (the content) to your AI website builder together. Part A says what to build and how it should look and behave. Part B holds the facts, tables and lists it must use exactly as written. Part C lists what is still missing or needs your confirmation. Real photographs will be added later, so the website must use dummy images for now (see B10).

The full list of research articles stays in your original Word file ("Description and information of Lab website"). Attach that file together with this one.

Contents

- **Part A. The prompt:** A1 Role and goal · A2 About the lab · A3 Design direction · A4 Site map and navigation · A5 Page-by-page requirements · A6 Lab Projects page behaviour · A7 Footer and contact form · A8 Technical requirements · A9 Content rules and working order
- **Part B. Content to use on the website:** B1 PI profile · B2 Lab projects table · B3 Technologies developed · B4 Training programmes · B5 Publications · B6 Datasets · B7 Lab members and alumni · B8 International collaborations · B9 Contact and social links
- **Part C. Items to confirm**

Part A. The prompt

Copy everything in Part A, and attach or paste Part B below it.

A1. Role and goal

You are a senior web designer and front-end developer who builds websites for leading research institutes. Build a complete, multi-page website for a virology and zoonotic-disease research laboratory led by **Dr. Ashwin Ashok Raut** at the **ICAR-National Institute of High Security Animal Diseases (NIHSAD), Bhopal, India**.

The website must do four things:

1. Present the lab's science clearly to scientists, funders, students and the public.
2. Show off the lab's rare high-containment facilities.
3. Give complete and accurate records of projects, publications, datasets, people and training.
4. Make it easy for visitors to propose a collaboration or apply for an open position.

A2. About the lab

- **Group:** Zoonotic Diseases, ICAR-NIHSAD, Bhopal, Madhya Pradesh, India.
- **Principal Investigator:** Dr. Ashwin Ashok Raut, Principal Scientist & Group Leader, Zoonotic Diseases.
- **Main work:** different types of viruses, mainly influenza viruses (avian influenza H5N1 in particular), together with other high-risk and emerging zoonotic viruses such as Nipah, MERS coronavirus, Ebola, Crimean-Congo haemorrhagic fever (CCHF), SARS-CoV-2 and Monkeypox.
- **The lab's four domains:**
  - Outbreak Investigation and Diagnostic Services
  - Disease Surveillance and Epidemiology
  - Infection Genomics
  - Next Generation Diagnostics and Vaccine Development
- **Rare facilities (feature these prominently across the site; photographs will be supplied later, so use dummy images for now):**
  - A **BSL-3 laboratory**.
  - An **Animal BSL-3 (ABSL-3) facility**.
  - A **Specific Pathogen Free (SPF) unit**.
  - **Large-animal experimental facilities** for pigs and cattle.
- **PI specialisation:**
  - Emerging Zoonotic Viral Diseases & One Health
  - Molecular Biology & Virology of Emerging Animal Diseases
  - Laboratory Biosafety and High-risk Biocontainment
  - Development of Next Generation Diagnostics & Vaccines
  - PathoGenomics, MetaViromics & Pathogen Discovery

A3. Design direction

Guiding themes

These four ideas inspire the whole website. Use them as the basis for the home-page tagline and for a strip of four short statements on the home page. Give the second theme the most visual weight.

1. **Pathogen discovery**
2. **Animal and human health** (very important: the One Health idea that animal, human and environmental health are connected)
3. **Searching the genomics**
4. **Translating the ideas of science to life**

Look and feel

- Clean, calm and professional, like the websites of well-established research labs. Content comes first.
- Restrained colour palette: deep navy or dark teal as the main colour, white and off-white backgrounds, and one muted accent colour. The light green and light pink-red tints are reserved for the Lab Projects table (see A6).
- Typography: a refined serif for headings and a clean sans-serif for body text. Comfortable line length, generous white space, and a strict grid.
- The lab will upload **real photographs** later (lab, facilities, team, fieldwork, Antarctica expedition). Until then, use **dummy images** in every photo slot, as described in B10, with a fixed aspect ratio so real images can replace them without changing the layout. Dummy images must be plain and neutral, never AI-generated pictures.
- Use simple, consistent line icons only where they help (for example the social-media links). Use charts or maps only where they carry real information.
- Fully responsive (phone, tablet, desktop). Meet WCAG AA contrast. Keyboard-accessible menus.
- Write the copy in plain, precise, scientific English. No hype and no marketing slogans.

Do not

- Do **not** use AI-generated images or anything that looks AI-made: glowing gradients, neon colours, 3D blobs, glassmorphism, floating-molecule stock art, robot or circuit imagery, or decorative emoji.
- Do not use stock illustrations of generic scientists or virus cartoons.
- Do not add animated effects that distract from reading. Subtle hover and scroll transitions are fine.
- Do not invent facts, numbers, publications or affiliations (see A9).

A4. Site map and navigation

Use exactly this order in the main menu. The menu is fixed to the top of the page while scrolling.

Menu item	Sub-sections	Notes
Home	None	See A5.
Research	Domains Technologies Developed Facilities (Advanced BSL-3 and SPF Unit)	Dropdown shows the sub-section names when the cursor hovers over “Research”.

Menu item	Sub-sections	Notes
Publications	Articles Book Chapters Patents	Dropdown on hover.
Datasets	Proteome Transcriptome Viral Metagenome	Dropdown on hover.
Lab Projects	None	Uses the whole project table (A6 and B2).
Lab Members	Current Alumni	Dropdown on hover.
Gallery	None	Photo gallery with categories.
News	Recent Achievements Lab Trainings Open Positions	Dropdown on hover. “Open Positions” is the website’s careers page.
Contact Us	None	Address plus the collaboration form.

Behaviour of the menu:

- Every menu item that has sub-sections shows their names in a dropdown when the cursor rests on it. Each name links to its own section or page.
- On touch screens, a tap opens the dropdown. On phones, the menu collapses into a clear mobile menu.
- The current page is highlighted. A breadcrumb or small page heading appears on inner pages.

A5. Page-by-page requirements

Home

1. **Hero:** lab name, a one-line statement built from the four guiding themes, a hero photograph (dummy image for now), and two buttons: “Explore our research” and “Collaborate with us”.
2. **Four guiding themes strip:** pathogen discovery, animal and human health, searching the genomics, translating science to life. One or two lines each.
3. **Meet the PI (required section):** photo, name, designation, a two-line summary, and a “Meet the PI” button. Clicking it opens the PI page (below).
4. **Lab at a glance:** short figures taken from B1 (projects, funding, articles, citations, datasets, trainings).
5. **Our four domains:** four cards linking to the Domains section.
6. **Rare facilities highlight:** BSL-3 laboratory, Animal BSL-3 facility, SPF unit, and pig and cattle facilities, each with a dummy image. Link to the Facilities page.
7. **Latest news:** the three most recent items, plus a visible “We are hiring” link to Open Positions when positions are open.

Meet the PI (opens from the home page)

This is a two-level experience. Clicking “Meet the PI” on the home page opens the PI page with the overview. On that page, clickable tabs or expandable blocks reveal further detail.

- **Level 1 (overview):** photo (dummy image for now), name, designation, specialisation, short biography, key figures. Use the profile in B1.

- **Level 2 (clickable tabs or expandable blocks): Education**, **Honors and Awards**, **Achievements**. The full content is in B1. Achievements are grouped by theme (diagnostic preparedness, biosafety and biocontainment, surveillance and epidemiology, diagnostics and vaccine development, infection genomics, human diseases, training).
- **Career timeline:** the positions table in B1, shown as a clean vertical timeline or table.
- **Professional recognitions** and **professional memberships:** from B1.
- **International collaborations and assignments:** a dedicated, dignified section on this page (content in B8). Show each partner with institution, country and topic. This is the agreed place for the lab's international collaborations.

Research

- **Domains:** four sections, one per domain. For each, write a short description in plain scientific language from the facts in A2 and B2, and list the related projects from the project table.
- **Technologies Developed:** cards for each technology in B3 (name, year, reference or repository code, and status). Show items marked "Under process for patenting" with that label. Add a second small group, "In development", for the ongoing-project items in B3.
- **Facilities (Advanced BSL-3 and SPF Unit):** a showpiece page with four sections: **BSL-3 laboratory**, **Animal BSL-3 (ABSL-3) facility**, **SPF unit**, and **Pig and cattle experimental facilities**. Each section has a title, one short plain sentence (for example, "Biosafety Level 3 laboratory for work with high-risk pathogens") and a small gallery of **three dummy images** (see B10). The lab will upload the real images later. Do not add technical specifications, capacities or claims that are not in Part B. If an optional "Key features" block is included, hide it until the lab fills it in; never show empty placeholders to visitors.

Publications

- A summary banner at the top: 60 research articles, cumulative impact factor above 170, more than 1200 citations, h-index 18, i10-index 30, and 5 book chapters (see C for the count to confirm).
- **Articles:** use the full "List of Research Publications" from the attached original Word file. Show each entry with authors (the PI's name in bold), title, journal, year, volume and pages, a DOI link, and the impact factor as given. Sort newest first. Add a year filter and a search box. A "Featured" row at the top may use the items in B5.
- **Book Chapters:** the list in B5.
- **Patents:** show the three technologies listed in B5 with the status "Under process for patenting" (no patent numbers yet). Build the entry layout (title, inventors, number, status, year) so that numbers can be added later.

Datasets

- Three sections: **Proteome**, **Transcriptome**, **Viral Metagenome** (content in B6).
- Show the opening line: the lab has published more than 100 high-throughput datasets, together with whole and partial genome sequences of viruses circulating in India.
- Each record shows accession number, description, database name and status. The accession number links to its database page (PRIDE, NCBI SRA, GEO or BioSample).
- Show a status label "Submitted, public release pending" for the viral metagenome records that are not yet public (see B6).

Lab Projects

See A6 for the exact behaviour. This page uses the whole table in B2.

Lab Members

- **Current:** two groups, **Research Team** and **Technical Staff** (names and roles in B7).
- Show members in a grid of **four per row**: first four in one row, next four in the next row, and so on. On tablets show two or three per row, and one on phones.
- Each card has a photo (neutral placeholder until supplied), name, designation, and a **three-line description**. Write plain placeholder descriptions for now; the lab will edit them later. Keep the descriptions in the editable data file, not in the page code.
- **Alumni:** all supervised and co-supervised research scholars from the supervision table in B7. Show name, degree, year, institution and dissertation title. Mark co-supervised scholars clearly.

Gallery

- A tidy image grid with a lightbox. Categories: Facilities, Fieldwork and Antarctica, Team and Events, Trainings.
- Every image has a caption. Use dummy images (B10) until the lab supplies photos.

News

- **Recent Achievements:** dated news items (title, date, short text, optional photo). Keep them in an editable data file. Seed with achievements from B1 (for example the 2026 publications and new National One Health Mission projects).
- **Lab Trainings:** show the 17 training programmes from B4 in a table, with a summary line (number of programmes and total trainees).
- **Open Positions:** this is the **careers page** of the website. List current openings (title, linked project, qualifications, how to apply). Projects marked “Positions open” on the Lab Projects page link here. Add a short note on how to apply and the contact email.

Contact Us

- **Main address** block at the top, with office phone, fax and both email addresses (B9).
- Below it, a section titled **“Looking for collaborations?”** with a form containing: Full name, Email address, “Tell us about yourself” (multi-line), and a **Submit** button.
- On submit, an email with the form contents must be sent to **ashwin.nihsad@gmail.com**. Show a clear success or error message and validate the fields (see A8 for how to make this work).

A6. Lab Projects page behaviour

Show the **whole table** in B2 with these columns: SN, Title of the project, Duration, Position, Funding Agency, Collaborating Institutions (number), Funds (in lakh INR). Put the summary statistics from B1 above the table. Do not add any extra columns.

- **Row colour:** ongoing projects have a **light green** background; completed projects have a **light pink-red** background. Add a small legend above the table. Colour the whole row.
- **How ongoing or completed is decided:** by the project’s end date compared with today’s date. Store a start and end date in the data file and calculate the status automatically, so rows change colour by themselves over time. The “Status” column in B2 shows the result as of October 2026 and is for checking only; it is **not** a visible column on the website.
- **Open positions marker:** for every ongoing project, add the words **“Positions open”** in **small font** at the **bottom right corner of the project-title cell**. Do not create a separate column for it. For now, treat all ongoing projects as having open positions.

- **The "Positions open" text is a clickable link** that goes to the Open Positions careers page inside this website (News > Open Positions).
- Optional helpers: filter buttons for All / Ongoing / Completed, and a search box. On phones, change the table into stacked cards that keep the same colours and the "Positions open" link.

A7. Footer and contact form

Every page ends with the same footer. It has two parts.

- **Contact and social row:** communication address, email addresses, and small clickable icons for **LinkedIn, Instagram, X (Twitter) and Facebook**. Use standard, recognisable social icons, small and uniform. Use the links in B9.
- **Quick links:** include the following (these are the important links, chosen for visitors):
  - Meet the PI
  - Research Domains
  - Facilities (BSL-3 and SPF Unit)
  - Publications
  - Datasets
  - Lab Projects
  - Lab Members
  - Open Positions
  - Lab Trainings
  - Collaborate with Us (Contact)
  - ICAR-NIHSAD official website (external link; URL to be added)
- A copyright line with the institute name and the year.

A8. Technical requirements

- Build a fast, secure, easy-to-host website. A static site generator or plain HTML, CSS and JavaScript is fine. Choose what keeps hosting and maintenance simple.
- **Editable content:** keep content in simple data files (JSON, CSV or Markdown) so a lab member with no coding skills can update them: projects, members, alumni, publications, news, training, open positions, gallery.
- **Contact form:** because static sites cannot send mail themselves, connect the form to a form-to-email service or a small serverless function so that submissions arrive at **ashwin.nihsad@gmail.com**. Add spam protection (a hidden honeypot field and basic rate limiting). Do not expose passwords or private keys in the page code.
- **Search-engine basics:** a title and description for every page, Open Graph tags, a sitemap, and structured data for the organisation and for articles.
- **Performance:** compress and lazy-load images; use modern image formats with sensible fallbacks.
- **Accessibility:** semantic HTML, alt text on every image, visible keyboard focus, and logical heading order.
- **Images:** create the dummy images as real image files in an images folder using the file names in B10, so the lab can replace each one by uploading a photo with the same name. Keep alt text and captions in the data files.
- Provide a short README that explains how to edit each data file, replace images, and publish the site.

A9. Content rules and working order

Content rules

- Use only the content in Part B and the attached publication list. Do not invent facts, figures, publications, patents, awards, affiliations or facility specifications.
- Where information is missing, show a clearly marked placeholder such as “[To be added]”. Never fill the gap with made-up text.
- Keep names, titles, numbers and dates exactly as in Part B.
- Highlight the PI’s name (Raut AA) in author lists.

Working order

1. Set up the design system (colours, fonts, grid, components) and the navigation with hover dropdowns.
2. Build Home and Meet the PI.
3. Build the data-heavy pages: Lab Projects, Publications, Datasets, Lab Members.
4. Build Research (Domains, Technologies, Facilities), Gallery, News and Contact.
5. Add the footer, test on phone, tablet and desktop, and test the contact form.
6. Hand over with the README and a list of every placeholder still to be filled.

Part B. Content to use on the website

Use this content exactly as written. Spellings in the original document have been corrected where they were clearly typing errors (see Part C).

B1. PI profile

Taken from the PI's full CV (April 2026). Date of birth, pay scale and school-level marks in the CV are deliberately not for the website.

Name	Dr. Ashwin Ashok Raut
Designation	Principal Scientist & Group Leader, Zoonotic Diseases
Institute	ICAR-National Institute of High Security Animal Diseases (ICAR-NIHSAD), Anand Nagar, Bhopal, Madhya Pradesh, India, 462022
Experience	Twenty years of research experience in molecular biology and virology, in different research positions through the Agricultural Research Service of ICAR.
Specialisation	Emerging Zoonotic Viral Diseases & One Health Molecular Biology & Virology of Emerging Animal Diseases Laboratory Biosafety and High-risk Biocontainment Development of Next Generation Diagnostics & Vaccines PathoGenomics, MetaViromics & Pathogen Discovery
Present work	Animal viral genomics and pathobiology of high-risk zoonotic viral diseases, particularly influenza. Leads the in-house team for diagnostic development and preparedness for multiple zoonotic diseases, and leads the NIHSAD team under the DBT Advanced Animal Disease Diagnosis and Management Consortium (ADMaC) for capacity building of the North-Eastern Region in animal disease diagnosis.
Professional memberships	Life member of: Indian Virological Society; Indian Association of Veterinary Public Health Specialists (IAVPHS), Izatnagar; Society for Biosafety, Bhopal; Society for Conservation of Domestic Animal Biodiversity (SOCDAB), Karnal; Association for the Promotion of DNA Fingerprinting and other DNA Technologies (ADNAT); Indian Society of Buffalo Development (ISBD), Hisar; Association of Avian Health Professionals (AAHP).

Education

Degree	Institution	Year	Notes
Post-Doctoral Fellow	Center of Excellence for Infectious Diseases, Texas Tech University Health Sciences Center, El Paso, Texas, USA	Jun 2009–Sep 2010	Worked on four projects (listed below).
PhD (Animal Biotechnology)	Indian Veterinary Research Institute (Deemed University), Izatnagar, UP	Feb 2006	First rank. Dissertation: "Evaluation of small interfering RNA (siRNA) as antiviral agent against rabies".
MVSc (Biotechnology)	Indian Veterinary Research Institute (Deemed University), Izatnagar, UP	Aug 1998	Dissertation: "Molecular Characterization of Rinderpest virus selected for thermostability".
BVSc & AH	Maharashtra Agriculture and Fisheries Sciences University (MAFSU), Nagpur	Dec 1995	Third rank. Gold Medal in Veterinary Pathology.

Post-doctoral projects:

- Development of siRNAi-based antiviral agent against neurotropic viral infections.
- Generation of a humanised mouse model to study novel therapies against HIV.
- Generation of a bioluminescent mouse model for acute myeloid leukemia to evaluate microRNA-based anti-cancer therapy.
- Designing of a microRNA-based multiple siRNA expression construct against avian influenza.

Career timeline

Position held	Place of work	From	To
Principal Scientist & Group Leader, Zoonotic Diseases	National Institute of High Security Animal Diseases (NIHSAD), Bhopal, MP	1 Nov 2014	Till date
Professor & HOD, Translational Medicine Dept.	All India Institute of Medical Sciences (AIIMS), Bhopal	1 May 2020	7 Dec 2020
Senior Scientist	National Institute of High Security Animal Diseases (NIHSAD), Bhopal, MP	1 Oct 2010	31 Oct 2014
BOYSCAST Fellow (Scientist on Deputation)	Texas Tech University Health Sciences Center, El Paso, Texas, USA	28 May 2009	28 Sep 2010
Scientist (Senior Scale)	Indian Veterinary Research Institute, Izatnagar, UP	6 Jul 2007	4 Apr 2009
Scientist	Central Institute for Research on Buffaloes, Hisar, Haryana, India	1 Sep 2005	5 Jul 2007
Scientist on Deputation	Indian Veterinary Research Institute, Izatnagar, UP	1 Sep 2002	31 Aug 2005
Scientist	Central Institute for Research on Buffaloes, Hisar, Haryana, India	1 Nov 1999	31 Aug 2002

Honors

- First Veterinarian, Animal Virology & Infectious Diseases Expert from India to visit Antarctica.
- BOYSCAST Fellowship 2009 from the Department of Science and Technology, Govt. of India, for overseas advanced research training for one year four months at Texas Tech University Health Sciences Center, El Paso, Texas, USA.
- Ranked first in PhD (Animal Biotechnology) at the Indian Veterinary Research Institute, Izatnagar, UP.
- Selected for the post of Agricultural Research Scientist by the national-level Agricultural Scientists Recruitment Board, Govt. of India.
- Qualified the National Eligibility Test in Animal Biotechnology in 1998.
- Junior Research Fellowship in Animal Biotechnology from the Indian Council of Agricultural Research (1996–98).
- Ranked third in BVSc & AH at MAFSU, Nagpur, and awarded the Gold Medal in Veterinary Pathology.
- National Merit Scholarship during higher secondary school and BVSc & AH (1989–95).

Awards

- Best Research Paper award, National Symposium and Annual Convention of the Indian Society for Buffalo Development, Bombay Veterinary College, 27–30 May 2006, for “Risks to animal health associated with artificial insemination using buffalo bull semen”.
- Best Research Paper award, Annual Conference of the Biotechnology Society, 2007, for “Production of recombinant adenoviruses expressing shRNAs targeting L and N genes of rabies virus”.

- Best Poster award, International Livestock and Dairy Expo (ILDEX-2008), New Delhi, for “PCR Assay for Detection of Buffalo Meat in Meat and Meat Products prepared under Different Manufacturing Conditions”.
- Best Poster award, Annual Symposium of the Indian Society for Veterinary Immunology and Biotechnology (ISVIB), IVRI, Izatnagar, 2013, for “Novel Molecular Beacon probe based real-time RT-PCR assay for diagnosis of Crimean-Congo Hemorrhagic Fever”.
- Best Poster award, Indian Poultry Science Association Conference (IPSACON 2024) and National Symposium “Shaping the Indian Poultry Sector for Sustainable Growth”, MAFSU-Nagpur Veterinary College, 16–18 October 2024, for “Identification of Potential Biomarkers of Highly Pathogenic Avian Influenza (H5N1) Infection Through Metabolomic Profiling in Chickens”.

Professional recognitions

- Technical Expert Committee, Livestock & Animal Sciences, DBT, MoST, GOI (2022–25)
- Technical Expert, Drafting of National Wildlife Health Policy, Central Zoo Authority, MoEFCC, GOI (2024)
- Expert Member, BSL4 Laboratory, ICMR-NIV, Pune (2022 onwards)
- Expert Member, ICMR Mobile BSL3 Laboratory Specifications Committee, DHR, GOI (2022–23)
- Domain Expert, Biobanking and ISO 20387:2018 Standards, Quality Council of India (2023)
- National Experts Stakeholders Committee, National Referral Centre for Wild Life, and One Health collaborations, Central Zoo Authority, MoEF, GOI (2023)
- Domain Expert, Strategy of Boosting Bio-Economy of MP, MP State Policy & Planning Commission (2023)
- Member, Committee for Establishment of an Institution like AIIMS in Veterinary Sciences, VCI (2023)
- Expert Member, Review Committee for Finalisation of Guidelines for Establishment of BSL3 Labs, DBT-ICMR, GOI (2020–23)
- Member, Research Advisory Committee, National Research Centre on Meat, Hyderabad (2018–21)
- DBT Expert Committee Member, Pathogen Discovery & One Health (2020–21)
- WHO Scientific TPP Review Committee Member, WHO R&D Blueprint Target Product Profile for Priority Diagnostics for Nipah virus (2019)
- Panel Member, Henipah & Ebolaviruses, BSL4 Network of Laboratories (BSL4ZNet), Canada (2018)
- Consultant for Design, Establishment & Commissioning of Clean-Air Animal Facility, IISER Bhopal (2014–18)
- Member Expert, Establishment of BSL2 & BSL3 facility at Assam Agricultural University, Guwahati (2014–20)
- Invited Member, ICMR Working Group on Zoonosis, Department of Health Research, GOI (2014–15)
- Invited Member, Task Force on Crimean-Congo Fever & Kyasanur Forest Disease, ICMR, DHR, GOI (2013–15)
- Panel Member, Schmallenberg Infection testing in bulls for import to India from Germany, NDP, GOI (2013)

Research summary and funding (for the Lab Projects page and the PI page)

- Carried out **32 research projects** (including 4 international and 19 multi-institutional collaborative projects): 16 as Principal Investigator and 16 as Co-Principal Investigator.
- **Extramural funded projects: 20** (7 as PI and 7 as Co-PI; 4 from DBT, 5 from ICAR, 5 from NOHM-ICMR). Four international funds: BBSRC (UK), NATP (World Bank), ASEAN, ILRI (Kenya).
- **Total extramural funding: more than ₹20 crore.**

Publication and data summary

- 60 research articles with a cumulative impact factor of more than 170 and more than 1200 citations to date, and 5 book chapters.
- Current h-index 18 and i10-index 30.
- Published partial and whole genome sequence data of several viruses circulating in India.
- Published more than 100 high-throughput datasets (transcriptomes, proteomes, metabolome, viral metagenomes and others).
- 17 training programmes conducted (see B4).

Achievements (from the CV; group on the PI page under these themes)

- **Diagnostic preparedness**
  - Established in-house diagnostic preparedness for CCHF, Nipah, MERS, Ebola, JE, West Nile fever, Hantaan, Rift Valley fever, Nairobi sheep disease and others. Involved in diagnosis and surveillance of these diseases at NIHSAD, the national referral laboratory for exotic and emerging animal diseases.
  - Adapted pan-family-wise and pan-genus-wise consensus PCR / RT-PCR assays for 12 viral families and 8 viral genera for zoonotic virus identification, recommended for pathogen discovery and outbreak investigation by PREDICT, USA.
  - Developed an in-house viral metagenomic pipeline for pan-viral identification, pathogen discovery, diagnosis and outbreak investigation on different high-throughput sequencing platforms.
  - Established protocols and a referral laboratory at the Central Institute for Research on Buffaloes, Hisar, as per OIE guidelines, for detection of pathogens transmitted through bubaline semen.
- **Biosafety and biocontainment**
  - Preliminary BSL-3 training at the Center of Excellence for Infectious Diseases, Texas Tech University Health Sciences Center, USA.
  - Ten years of working in the BSL-3 biocontainment laboratory of NIHSAD, involved in laboratory management in various capacities.
  - Provides training and capacity building in biosafety and biosecurity to workers and researchers from India and abroad.
  - Biosafety advisor for the Indian Scientific Expedition to Antarctica.
  - Consultant for the establishment of the clean-air animal facility at IISER Bhopal.
- **Surveillance and epidemiology**
  - Identified mixed infection of classical swine fever and Japanese encephalitis in pigs and its putative role in the current wave of JE in humans in North-Eastern India.
  - Investigated the first human Crimean-Congo haemorrhagic fever outbreak (2011, Gujarat) and traced cattle and Hyalomma anatolicum ticks as reservoir hosts; continued surveillance in the outbreak areas.
  - Investigated putative reservoir host animals after the Nipah outbreaks of 2018 and 2019 in Kerala.
  - Conducted surveillance of MERS-CoV in the Indian camel population.
  - Isolated influenza A viruses from migratory ruddy shelducks and bar-headed geese.
  - Isolated emerging pigeon paramyxovirus-1 (PPMV-1) from pigeons in Madhya Pradesh.
  - Used metaviromics to establish the exchange of viromes at the interface of migratory and native aquatic birds during the migratory season in North-Eastern India.
  - Generated virus profiles of the Antarctic Adélie penguin, South Polar skua, Weddell seal and ornithogenic soil.

- **Diagnostics and vaccine development**
  - Prototype molecular-beacon diagnostic PCR array for simultaneous detection of 17 emerging viruses (CCHFV, JEV, VSV, Aujeszky's, RVFV, WNFV, Ebola, Marburg, Hantaan, EEEV, Schmallenberg, BVDV, CAEV, ASFV, SVDV, TGEV, Nipah).
  - Prototype triplex nucleic acid lateral flow immunoassay (NALFIA) for strip-based detection of Nipah, Ebola and MERS at the same time.
  - Novel molecular-beacon diagnostic test for rapid detection of CCHFV.
  - Expressed recombinant antigens of CCHF, MERS, Ebola, NSD and Lyssa viruses in a prokaryotic system, and produced hyperimmune sera for Nipah, Reston Ebola and MERS viruses.
  - Developed a thermoresistant rinderpest vaccine and a canine parvovirus DNA vaccine.
- **Infection genomics**
  - Identified differences in pathway activation and effector molecules in response to HPAIV H5N1 between the susceptible chicken and the tolerant duck, using genome-wide transcriptomics and proteomics.
  - Identified the role of SOCS1, SOCS3, IFNAR1 and IFNAR2 in pathogenesis and mortality of crows due to H5N1 infection.
  - Identified miR-485, which targets the PB1 gene of avian influenza virus and the host RIG-I, as a novel anti-influenza therapeutic molecule.
  - First whole genome sequences from Indian isolates of swinepox, duck plague, varicella-zoster, Japanese encephalitis genotype I and influenza A (H3N8).
- **Human diseases**
  - Generated a humanised mouse model (human fetal liver and thymus implantation and cord-blood CD34+ stem cells in NOD/SCID/gc-/- neonatal mice) to evaluate novel HIV therapies.
  - Generated a bioluminescent mouse model for acute myeloid leukemia to evaluate microRNA-based anti-cancer therapy.
  - Screened native Indian tribes for genetic markers of resistance or susceptibility to tuberculosis and HIV.
- **Training and human resource development**
  - Course Coordinator of the ASEAN-India Fund International Training-cum-Workshop on diagnosis of transboundary animal diseases, attended by 13 participants from seven ASEAN countries.
  - Course Coordinator of more than 12 training programmes in the last 5 years on biosafety and diagnosis of emerging zoonotic diseases.
  - In-charge of the institute's HRD Cell for the past 6 years.
  - Guided 5 PhD, 5 MVSc, 2 BS-MS and 2 MSc students in the molecular biology of viral diseases.

B2. Lab projects table (use the whole table)

Funds are in lakh INR. "Status" is the status as of October 2026 and is given here only so the builder can check its automatic colouring. Ongoing = light green row with the small "Positions open" link. Completed = light pink-red row.

SN	Title of the project	Duration	Position	Funding agency	Collab. inst.	Funds (lakh INR)	Status
1	Establishing a One Health Sentinel Surveillance System in High Human–Animal Interaction Settings among Selected Zoos in India. (Collaboration of 18 institutes)	2026–2029	PI	National One Health Mission & ICMR	18	265.51	Ongoing

SN	Title of the project	Duration	Position	Funding agency	Collab. inst.	Funds (lakh INR)	Status
2	A One Health Approach to National Pandemic Preparedness: Development of a Novel Multiplex Assay for Serological Investigations of Zoonotic Viruses in India. (Collaboration with IISER, Bhopal & NCCS, Pune)	2026–2029	PI	National One Health Mission & ICMR	5	146.15	Ongoing
3	Preclinical development of neutralizing novel human monoclonal antibodies against Rabies for Post-exposure prophylaxis. (Collaboration with NCCS, Pune & IISER, Bhopal)	2026–2029	Co-PI	National One Health Mission & ICMR	4	65.00	Ongoing
4	NOHM-Building a surveillance model for detecting zoonotic spillover in increased bird-human interaction settings using the One Health Approach: A study at selected bird sanctuaries and wetlands.	2025–2027	PI	National One Health Mission & ICMR	10	275.41	Ongoing
5	NOHM-National network of BSL3/4 Laboratories	2024 (started)	Co-PI	National One Health Mission & ICMR	24	49.60 per year	Ongoing
6	NOHM-Training cum Workshop on Biosafety for handling and diagnosis of high-risk animal pathogens in ABSL3/BSL3 Laboratories	2024–2025	PC	National One Health Mission & ICMR	24	78.94	Completed
7	ICAR-AINP-One Health: Subproject: Surveillance of Nipah, MERS, Ebola viruses in bat populations.	2024–2026	PI	ICAR-All India Network Project-One Health	15 (IVRI)	65.00	Ongoing
8	NP-GET-Subproject: Development of gene edited chicken with enhanced resilience to avian influenza (H5N1)	2024–2026	Co-PI	ICAR-All India Network Project-Gene Editing	2 (CARI)	70.00	Ongoing
9	Investigation of HPAIV (H5N1) Pathogenesis in chicken by metabolomics approach	Apr 2024–Mar 2026	Co-PI	Intramural Institute Project	1	~20.00	Completed
10	Surveillance of rodent-borne pathogens in synanthropic rats associated with Musahari community of North India.	Apr 2024–Mar 2026	Co-PI	Intramural Institute Project	1	~20.00	Completed
11	Development of vaccines for prioritized exotic & emerging animal diseases. (ICAR-NIHSAD & CGIAR-ILRI Collaboration)	2023–2027	PI	ICAR-International Livestock Res. Institute	2	~100.00	Ongoing
12	Studies on host pathogen interactions & development of vaccine for zoonotic coronaviruses.	Jun 2021–May 2024	Co-PI	National Agricultural Science Fund, ICAR	3	262.10	Completed
13	Epidemiological studies & development of antiviral therapeutics against coronaviruses	Jun 2021–May 2024	PI	National Agricultural Science Fund, ICAR	3	85.32	Completed
14	Development and validation of LFA POCT Kits for SARS-CoV-2 antigen detection (BIRAC)-In Collaboration with DeNovo Bio Labs	2020–2021	Co-PI	DBT-BIRAC	2	18.90	Completed
15	DBT-NER-Advanced Animal Disease Diagnosis and Management Consortium (ADMaC)	Jun 2014–Oct 2019	PC & PI	Dept of Biotechnology, GOI	15	211.49	Completed
16	Identification of the molecular basis of differential host responses to rapidly evolving Avian Influenza viruses in different avian species.	Feb 2014–Jan 2018	Co-PI	BBSRC, UK and DBT, India	2	111.63	Completed
17	Investigation of Antarctic Animal Metavirome: An initiative for Pathogen Discovery with special reference to globally emerging avian influenza and other high risks viruses.	Apr 2017–Mar 2020	PI	Ministry of Earth Sciences, Govt of India	2	82.00	Completed

SN	Title of the project	Duration	Position	Funding agency	Collab. inst.	Funds (lakh INR)	Status
18	Viral Metagenomic Profiling of Native and Migratory Aquatic Birds of North-Eastern States to Unravel Influenza Virus Ecology Harboured in them.	Sep 2014–Aug 2017	PI	Dept of Biotechnology, Govt of India	2	84.20	Completed
19	Surveillance of Exotic and Emerging Animal Diseases in Indian and Imported Livestock & Poultry and Their Products. (Service Project) (Diagnostic Services for High-risk Zoonotic Diseases: CCHF, Nipah, MERS, COVID-19, KFD, RVF, NSD, JE, WN, PRV, Monkeypox etc.)	2014 onwards (continuous)	Co-PI	ICAR-NIHSAD, Bhopal	1	--	Ongoing
20	Investigation of Schmallenberg & related viruses in Ruminants and Culicoides spp.	Apr 2015–Mar 2018	Co-PI	ICAR-NIHSAD	1	10.00	Completed
21	Consortium Research Platform on Vaccines and Diagnostics: “Development of DNA strip test for sub-typing of avian influenza viruses”.	Aug 2015–Jul 2018	Co-PI	CRP, ICAR	1	33.60	Completed
22	Development of whole genome sequencing protocols for viruses on Nanopore sequencing platform	Apr 2017–Mar 2020	Co-PI	ICAR-NIHSAD, Bhopal	1	18.00	Completed
23	ASEAN-India Fund project on Training cum Workshop on Conventional and Molecular Techniques for diagnosis of transboundary animal diseases.	2013–2014	PC	ASEAN-India Collaboration Fund	7 countries	~50.00 (USD 77,506)	Completed
24	Quest for a Universal Diagnostic PCR Array for multiple exotic and emerging viral diseases.	Jun 2012–Jul 2014	PI	ICAR-IVRI, Izatnagar	1	20.00	Completed
25	Development of Real-Time PCR for diagnosis of Crimean Congo Hemorrhagic Fever.	Apr 2011–Mar 2012	PI	ICAR-IVRI, Izatnagar	1	10.20	Completed
26	Evaluation of Lentiviral Vector based siRNA delivery system for antiviral RNA interference in poultry.	Jul 2008–Jun 2011	PI	ICAR-IVRI, Izatnagar	1	17.52	Completed
27	Identification of miRNAs expressed in bubaline mammary tissue.	Jul 2008–Jun 2011	Co-PI	ICAR-IVRI, Izatnagar	1	18.90	Completed
28	Determination of phylogeographical dynamics and biodiversity of Indian buffalo breeds using mitochondrial DNA analysis.	Mar 2006–Mar 2008	PI	ICAR-CIRB, Hisar	1	12.00	Completed
29	Semen Quality Testing Laboratory, Network Project on Buffaloes. Service Project: Detection of pathogens transmitting through semen	Apr 2001–Aug 2002	PI & Incharge	ICAR-Network on Buffaloes	1	25.00	Completed
30	Monitoring and surveillance of infectious diseases associated with mortality in buffalo calves.	Mar 2000–Aug 2002	Co-PI	ICAR-CIRB, Hisar	1	15.50	Completed
31	Identification of molecular markers applicable for progeny testing in buffaloes. (World Bank funded National Agricultural Technology Project under Competitive Grant Program).	Jun 2001–Aug 2002	Co-PI	World Bank funded ICAR-National Agricultural Technology Project	1	30.00	Completed
32	Identification of molecular markers and marker assisted selection for improvement of production in buffaloes.	Mar 2000–May 2001	Co-PI	ICAR-CIRB, Hisar	1	10.00	Completed

Position codes: PI = Principal Investigator; Co-PI = Co-Principal Investigator; PC = Programme / Course Coordinator.

B3. Technologies developed

Taken from the “Technologies Developed” section of the PI’s CV. Items 8 to 10 are under process for patenting.

#	Technology	Reference or status	Year
1	Triplex Nucleic Acid Lateral Flow Immunoassay (NALFIA) test for strip-based simultaneous detection of Nipah, Reston Ebola and MERS viruses	Animal Diseases (BMC). doi.org/10.1186/s44149-024-00127-w	2024
2	Multiplex Real-Time RT-PCR kit for Avian Influenza A virus typing and H5 and H9 subtyping	Technology Code 201596626809418 (ICAR-KRISHI Technology Repository)	2021
3	Multiplex Real-Time RT-PCR kit for Avian Influenza A virus typing and H5N1 subtyping	Technology Code 201596857548890 (ICAR-KRISHI Technology Repository)	2021
4	Novel molecular-beacon probe-based rapid diagnostic test for Crimean-Congo Hemorrhagic Fever virus detection	Biomedical Research International. DOI 10.1155/2014/496219	2014
5	MicroRNA-485 as a novel therapeutic molecule for control of influenza infection	Science Signaling. DOI 10.1126/scisignal.aab3183	2015
6	Upgradation of the commercially available 44K to a 60K chicken microarray chip for comprehensive whole genome expression profiling	PLoS One. DOI 10.1371/journal.pone.0153671	2016
7	Design of a 60K duck-specific microarray chip for whole genome expression profiling	Acta Virologica. DOI 10.4149/av_2017_01_66	2017
8	SYBR Green chemistry-based universal diagnostic real-time PCR array for simultaneous detection of 22 exotic and emerging viruses	Under process for patenting	—
9	Molecular-beacon chemistry-based universal diagnostic real-time PCR array for simultaneous detection of 17 exotic and emerging viruses	Under process for patenting	—
10	Artificial positive control construct for molecular diagnosis of 16 tick-borne viruses and 12 porcine viruses	Under process for patenting	—
11	Thermoresistant rinderpest vaccine	Acta Virologica 45:235–241	2001
12	Canine parvovirus DNA vaccine	Current Science 88(5):778–781	2004
13	Recombinant soluble glycoprotein antigen for rabies diagnosis	Veterinary Microbiology 108:207–214	2005

In development (from current projects)

Show in a separate “In development” group on the Technologies page.

#	Technology	Project
1	Novel multiplex assay for serological investigation of zoonotic viruses in India	Project 2 (PI), with IISER Bhopal and NCCS Pune; NOHM-ICMR
2	Neutralising human monoclonal antibodies against rabies for post-exposure prophylaxis (preclinical)	Project 3 (Co-PI), with NCCS Pune and IISER Bhopal; NOHM-ICMR
3	Gene-edited chicken with enhanced resilience to avian influenza (H5N1)	Project 8 (Co-PI); ICAR Network Project on Gene Editing
4	Vaccines for prioritised exotic and emerging animal diseases	Project 11 (PI); ICAR-NIHSAD and CGIAR-ILRI

B4. Training programmes or refresher courses conducted

Show under News > Lab Trainings. Total: 17 programmes. “Course Coordinator” = the PI coordinated the course; “Core Faculty” = the PI taught on the course.

#	Training (title)	Duration / year	Beneficiaries	Role
1	ASEAN-India Cooperation Fund Sponsored International Training cum Workshop on “Conventional and Molecular Techniques for diagnosis of Transboundary Animal Diseases”	7 days (18–25 Feb 2014)	7	Course Coordinator
2	ASEAN-India Cooperation Fund Sponsored International Training cum Workshop on “Conventional and Molecular Techniques for diagnosis of Transboundary Animal Diseases”	7 days (28 Feb–7 Mar 2014)	6	Course Coordinator

#	Training (title)	Duration / year	Beneficiaries	Role
3	Laboratory Biosafety and Biosecurity for Handling Transboundary Animal Pathogens under DBT-NER-ADMaC Consortium	7 days (25 Feb–3 Mar 2015)	5	Course Coordinator
4	"Biosafety and Biosecurity for handling and identification of zoonotic pathogens" organized under DBT-NER-ADMaC Consortium	5 days (16–20 Feb 2016)	10	Course Coordinator
5	Biosafety & Biosecurity for handling & identification of Emerging Zoonotic Pathogens	5 days (18–22 Jul 2016)	8	Course Coordinator
6	Biosafety & Biosecurity for handling & identification of Emerging Zoonotic Pathogens	5 days (3–8 Aug 2016)	8	Course Coordinator
7	Biosafety & Biosecurity for handling & identification of Emerging Zoonotic Pathogens	5 days (3–7 Oct 2016)	5	Course Coordinator
8	Biosafety & Biosecurity for handling & identification of Emerging Zoonotic Pathogens	5 days (17–21 Jan 2017)	7	Course Coordinator
9	Diagnostic Techniques for Emerging Animal Diseases	5 days (3–8 Aug 2017)	10	Course Coordinator
10	Hands-on Training on Emerging Animal Disease Diagnosis, under ADMaC	8 days (1–8 Aug 2018)	8	Course Coordinator
11	Training on Biosafety and Diagnosis of Emerging Animal Disease, under ADMaC	8 days (5–11 Mar 2019)	6	Course Coordinator
12	Training on Laboratory Biorisk Management and Investigation of Emerging Animal & Zoonotic Diseases with One Health Approach	8 days (1–8 Aug 2019)	9	Course Coordinator
13	PCR in disease diagnosis, Gene detection and species differentiation, conducted under Niche Area of Excellence	7 days (2007)	22	Core Faculty
14	Gene expression techniques: Gene amplification, cloning, characterization of recombinant clones and in vitro expression analysis, DBT sponsored training program	21 days (2007)	20	Core Faculty
15	Bioinformatics as a tool for gene amplification, cloning and molecular modeling, conducted under Niche Area of Excellence	7 days (2008)	45	Core Faculty
16	Applications of molecular methods in livestock disease diagnosis and research, conducted under DBT Project at Dept of Veterinary Microbiology and Biotechnology, Nagpur Veterinary College, Nagpur	21 days (2008)	19	Core Faculty
17	Advanced Molecular Biology Tools used in Animal Disease Research and Diagnosis, DBT sponsored training program	21 days (2008)	18	Core Faculty

B5. Publications

Articles

Use the "List of Research Publications" in the attached original Word file. It has the full citation, DOI and impact factor for every article. Impact factors are shown as given in the original (older entries follow the 2021 update; recent entries show the 5-year impact factor).

Suggested "Featured" row (PI is a co-author on each):

Year	Publication	Journal	Impact factor
2021	Mishra A et al. SARS-CoV-2 Delta Variant among Asiatic Lions, India. 27(10):2723–2725.	Emerging Infectious Diseases	16.126
2020	Mishra R et al. Particulate matter (PM10) enhances RNA virus infection through modulation of innate immune responses. 266:115148.	Environmental Pollution	9.988
2022	Kumar A et al. Essential role of Rnd1 in innate immunity during viral and bacterial infections. 13(6):520.	Cell Death & Disease	9.685

Year	Publication	Journal	Impact factor
2015	Ingle H et al. The microRNA miR-485 targets host and influenza virus transcripts to regulate antiviral immunity and restrict viral replication. 8(406):ra126.	Science Signaling	9.517
2022	Mishra A et al. Differential miRNA expression profiling of highly pathogenic avian influenza virus H5N1 infected chicken lungs.	Virologica Sinica	6.947
2018	Kumar A et al. MicroRNA hsa-miR-324-5p suppresses H5N1 virus replication by targeting the viral PB1 and host CUEDC2. 92:e01057-18.	Journal of Virology	6.549
2026	Vijaykumar P et al. Systems-level analysis of HPAI H5N1 infection in ducks: integrating transcriptomic, proteomic and phosphoproteomic data. 27, 2884.	Int. J. Molecular Sciences	5.7 (5-yr)
2024	Chingtham S et al. Novel triplex nucleic acid lateral flow immunoassay for rapid detection of Nipah virus, MERS coronavirus and Reston ebolavirus. 4, 21.	Animal Diseases	0.742

Book chapters

1. Mishra A and **Raut AA** (2012). Nutrigenomics & livestock productivity. In: Animal Nutrition: Advances & Developments. Satish Serial Publishing House, Lucknow. pp 164–166.
2. Kulkarni DD, Tosh C, Bhatia S, **Raut AA** (2017). Nipah Virus Infection. In: Bayry J (ed) Emerging and Re-emerging Infectious Diseases of Livestock. Springer, Cham. https://doi.org/10.1007/978-3-319-47426-7_12
3. Sudhakar SB, Gandhale PN, Singh F, **Raut AA**, Mishra A, Kulkarni DD, Singh VP (2020). Schmallenberg Virus. In: Malik Y, Singh R, Yadav M (eds) Emerging and Transboundary Animal Viruses. Livestock Diseases and Management. Springer, Singapore. https://doi.org/10.1007/978-981-15-0402-0_10
4. **Raut AA**, Gandhale PN, Kumar N, Pateriya A, Mishra A, Kulkarni DD (2020). Crimean-Congo Haemorrhagic Fever Virus. In: Malik Y, Singh R, Yadav M (eds) Emerging and Transboundary Animal Viruses. Springer, Singapore. https://doi.org/10.1007/978-981-15-0402-0_11
5. **Raut AA** (2024). Monkeypox Virus. In: Status of Emerging Viral Diseases of Animals in India, edited by Bhatia S, Tosh C, Mishra N, Raut AA, Rajukumar K, Sanyal A. ICAR-National Institute of High Security Animal Diseases. pp 108–112.
6. **Raut AA**, Mishra A, Pathak A (2024). Ebola Virus. In: Status of Emerging Viral Diseases of Animals in India (as above). pp 87–93.
7. **Raut AA**, Mishra A, Pathak A (2024). Middle East Respiratory Syndrome Coronavirus. In: Status of Emerging Viral Diseases of Animals in India (as above). pp 103–107.

Patents

No patent numbers have been granted or filed yet in the CV. Show these three as “Under process for patenting”:

1. SYBR Green chemistry-based universal diagnostic real-time PCR array for simultaneous detection of 22 exotic and emerging viruses.
2. Molecular-beacon chemistry-based universal diagnostic real-time PCR array for simultaneous detection of 17 exotic and emerging viruses.
3. Artificial positive control construct for molecular diagnosis of 16 tick-borne viruses and 12 porcine viruses.

B6. Datasets (high-throughput data in the public domain)

Proteome

Accession	Description	Database
PXD010358	Lung proteome of highly pathogenic avian influenza virus (H5N1) infection progression in chicken	ProteomeXchange Consortium via PRIDE (proteomecentral.proteomexchange.org)

Transcriptome

Accession	Description	Database
SRS498120, SRS497174	Lung transcriptome of non-infected and highly pathogenic avian influenza virus infected crow on 454 GS FLX Titanium	NCBI SRA
GSE65230 (GSM159070–77)	Duck gene expression profile by microarray on Agilent (8 samples)	NCBI GEO
GSE65231 (GSM159078–85)	Chicken gene expression profile by microarray on Agilent (8 samples)	NCBI GEO

Viral Metagenome (Illumina platform)

The first record is public. The remaining records are submitted and will be released to the public next year (show "Submitted, public release pending").

Accession	Description	Database
SRX1444196	Profiling of gut virome of domestic ducks	NCBI SRA
SRR9112925–28	Pre-migratory oral profile of native and migratory aquatic birds from fresh water lakes in Assam state	NCBI SRA
SRR9112931–34	Post-migratory oral profile of native and migratory aquatic birds from fresh water lakes in North Eastern India	NCBI SRA
SRR9112944–47	Pre-migratory cloacal profile of native and migratory aquatic birds from fresh water lakes in Assam state	NCBI SRA
SRR9112992–93	Post-migratory cloacal profile of native and migratory aquatic birds from fresh water lakes in Assam state	NCBI SRA
SAMN1112706	Adélie penguin oral viral metagenome from Svenner Islands, Antarctica, Jan–Feb 2018	NCBI BioSample
SAMN11119281	South Polar Skua faecal viral metagenome from near Bharati Indian Research Station, Prydz Bay, Antarctica, Jan–Feb 2018	NCBI BioSample
SAMN11127047	Weddell Seal faecal viral metagenome from near Bharati Indian Research Station, Prydz Bay, Antarctica, Jan–Feb 2018	NCBI BioSample
SAMN11126887	Adélie penguin faecal viral metagenome from Svenner Islands, Antarctica, Jan–Feb 2018	NCBI BioSample
SAMN11127117	Antarctic ornithogenic soil viral metagenome from Svenner Islands, Antarctica, Jan–Feb 2018	NCBI BioSample
SAMN11127116	Antarctic freshwater lake viral metagenome from Fisher Island, Antarctica, Jan–Feb 2018	NCBI BioSample

B7. Lab members and alumni

Current team: Research Team

Show in a grid of four per row. Each card: photo, name, designation, three-line description (placeholder text for now).

Name	Designation
Dr. Anamika Mishra	Principal Scientist at ICAR-NIHSAD
Dr. Piyus Joshi	Project Research Scientist-III
Dr. Shailendar Mishra	Project Research Scientist-II
Dr. Debabrata Dandsena	Project Research Scientist-II
Dr. Sonal Sharma	Research Scholar (PhD)
Varsha Rani Agrawal	MVSc Scholar, Veterinary Public Health
Vedansh Trivedi	Research Scholar (PhD)
Deepak	Project Fellow
Prince Kumar	ILRI (International Livestock Research Institute) Consultant for MERS vaccine
Himanshu	Project Research Scientist-I (MMSA Project)
Muskan	Project Fellow
Dr. Arti Chaurasiya	Project Fellow
Dr. Piyush Soni	Project Fellow

Current team: Technical Staff

Surendar · Bharat · Ankit · Bharat · Jagdish · Aran (see Part C about the repeated name and spelling).

Alumni: research scholars supervised

#	Name	Title of dissertation	Degree	Year
1	Rupam Dutta (IVRI, Izatnagar)	Identification of the Dynamics of Viral Metagenome between Native and Migratory Duck and Goose in North-Eastern Region of India	PhD	2017
2	Chandan K (IVRI, Izatnagar)	Development of Multiplex Nucleic Acid Lateral Flow Assay for Hemagglutinin (HA) subtyping of Avian Influenza Virus	MVSc	2017
3	Mohammed Fawaz (IVRI, Izatnagar)	Metagenomic profiling of gut virome of domestic ducks	MVSc	2014
4	Aman Kamboj (IVRI, Izatnagar)	Evaluation of real time RT-PCR for detection of CCHF virus	MVSc	2012
5	Ravi Prakash (IVRI, Izatnagar)	Construction of artificial miRNA polycistron encoding multiple anti-avian influenza virus siRNAs	MVSc	2010
6	Basavaraj Sajjanar (IVRI, Izatnagar)	Identification of small interfering RNAs for silencing of Infectious Bursal Disease Virus infection	MVSc	2008
7	Sachin Samuchiwal (GJUST, Hisar)	Identification and screening of stem cell markers in umbilical cord blood of buffalo	MSc	2008
8	Sandeep Kumar (GJUST, Hisar)	Molecular characterization of exon-8 of ovine DGAT-1 gene	MSc	2007
9	Anil Kumar (GJUST, Hisar)	Molecular characterization of major candidate genes associated with milk fat percentage in Indian buffalo	PhD	2006

Alumni: as co-supervisor

#	Name	Title of dissertation	Degree	Year
10	Santhalembi Chingtham (IVRI, Izatnagar)	Development of Multiplex Diagnostic Assay for Nipah Virus, Reston Ebolavirus, and Middle East Respiratory Syndrome Coronavirus infections in Bats	PhD	2018

#	Name	Title of dissertation	Degree	Year
11	P Vijaykumar (IVRI, Izatnagar)	Lung proteome analysis of Highly pathogenic avian influenza virus H5N1 infection progression in chicken and duck	PhD	2016
12	Pradeep Ranaware (IVRI, Izatnagar)	Experimental transmission of highly pathogenic duck isolate of H5N1 and host gene expression	PhD	2014

B8. International collaborations and assignments

Show these on the Meet the PI page as a clean list or small grid (institution, country, topic).

Partner	Country	Topic or role
Dr Dave Burt, Roslin Institute, University of Edinburgh	United Kingdom	Joint research programme on avian influenza infection genomics, funded by BBSRC (UK) and DBT (India)
Australian Animal Health Laboratory, Geelong	Australia	Nipah diagnostics
Dr Bart Haagmans, Viroscience Lab, Erasmus	Netherlands	MERS pathobiology
Dr Branka Horvat, INSERM, Lyon	France	Nipah virus–bat interplay
Import testing panel (Germany to India)	India / Germany	Panel member for testing of Schmallenberg infection in bulls selected for import to India from Germany
BSL4ZNet network of laboratories	International	Member of the international network

B9. Contact and social links

Main address	ICAR-National Institute of High Security Animal Diseases (ICAR-NIHSAD) Anand Nagar, Bhopal, Madhya Pradesh, India, 462022
Office phone	+91-755-2750647 (ext. 398)
Fax	+91-755-2758842
Email (collaboration form is sent here)	ashwin.nihsad@gmail.com
Official email	ashwin.raut@icar.gov.in
LinkedIn	[URL to be added]
Instagram	[URL to be added]
X (Twitter)	[URL to be added]
Facebook	[URL to be added]

The PI's mobile number appears in the CV. Do not publish it unless the PI confirms.

B10. Image slots (use dummy images for now)

The lab will upload real photographs later. Create every slot below as a real image file with a plain **dummy image**: a neutral light-grey frame with a thin border, a small camera outline icon in the centre, and the caption "Photo coming soon". Do not use AI-generated pictures, stock photos of people or invented scenes. Use the file names below so that the lab can replace each image by uploading a photo with the same name.

Image slot	Ratio	Used on and file name
Home hero	16:9	Home. images/home/hero.jpg

Image slot	Ratio	Used on and file name
PI portrait	4:5	Home (Meet the PI) and PI page. images/pi/portrait.jpg
BSL-3 laboratory (3 images)	3:2	Facilities and home highlight. images/facilities/bsl3-01.jpg to bsl3-03.jpg
Animal BSL-3 facility (3 images)	3:2	Facilities and home highlight. images/facilities/abs3-01.jpg to abs3-03.jpg
SPF unit (3 images)	3:2	Facilities and home highlight. images/facilities/spf-01.jpg to spf-03.jpg
Pig and cattle facilities (2 images)	3:2	Facilities. images/facilities/large-animal-01.jpg and large-animal-02.jpg
Gallery (3 per category, 4 categories)	4:3	Gallery. images/gallery/facilities-01.jpg … trainings-03.jpg
Team member default	1:1	Lab Members. images/members/default.jpg (neutral silhouette)
News default	16:9	News items without a photo. images/news/default.jpg

Part C. Items to confirm

These are the gaps and judgement calls found while organising your description. None of them stops the website from being built; each shows as a marked placeholder or follows the assumption given until you reply.

#	Item	What is needed or what was assumed	Where it shows
1	Lab name, logo, tagline	Not given. The working title is “Zoonotic Diseases Group, ICAR-NIHSAD”.	Header, home
2	Social-media URLs	LinkedIn, Instagram, X and Facebook links are marked “[URL to be added]”.	Footer, contact
3	PI photo	The CV contains a portrait, but it could not be placed into this prompt. Please send the photo as a separate JPG or PNG. A dummy image is used until then.	Home, PI page
4	Real photographs	Dummy images are used for the hero, PI, BSL-3, Animal BSL-3, SPF unit, pig and cattle facilities and gallery. Replace them using the file names in B10.	Throughout
5	Facility details	The SPF egg-sourcing detail was removed as you asked. Only short, factual descriptions are used. Capacities, specifications and certifications are not shown until you supply them.	Facilities
6	Personal details left out	Date of birth, pay scale and school-level marks in the CV are not used. The PI’s mobile number is shown only if the PI confirms.	PI page, contact
7	Patents	The CV lists three technologies “under process for patenting” and no patent numbers. They are shown with that status.	Publications, Technologies
8	Publication list	Not reproduced here. Attach the original Word file (or the CV) with this document.	Publications
9	Book chapter count	The summary says 5 chapters, but 7 are listed. All 7 are shown. Confirm the figure to display.	Publications
10	Extramural funding counts	The summary gives 20 funded projects but lists 7 as PI and 7 as Co-PI (14). Shown as written. Please check.	Lab Projects, PI page
11	Projects 7 and 8 (end 2026) and 9 and 10 (end Mar 2026)	As of October 2026, 7 and 8 are treated as ongoing (the year has not ended); 9 and 10 are treated as completed.	Lab Projects
12	“Positions open”	All ongoing projects carry the link for now. Tell us which projects have real openings, and add the openings to the careers page.	Lab Projects, News
13	Technologies Developed	Now taken from the CV (13 items). The four “In development” items come from your current projects. Please confirm.	Research
14	Technical staff	“Bharat” appears twice and “Aran” may be a different spelling. Please confirm names and add designations.	Lab Members
15	Team descriptions	Three-line descriptions are placeholders for you to edit.	Lab Members
16	Viral metagenome release status	The original says records “ii onwards” are submitted but not yet public. We read this as everything after the first record.	Datasets
17	Accession SAMN1112706	It has fewer digits than the other BioSample IDs. Please check it.	Datasets
18	Menu hover wording	You wrote that “two sections name should hover”. We applied the hover dropdown to every menu with sub-sections. Tell us if you wanted it only for Research.	Navigation

#	Item	What is needed or what was assumed	Where it shows
19	Spelling and wording changes	Adélie penguin (was “Aedile”); South Polar Skua (was “Skuva”); Edinburgh; Quality Council of India (CV says “Quality control of India”); “transboundary”; dates written in full. The CV line “Duck Papillomo virus Genotype VI” was left out because it may mean pigeon paramyxovirus-1 genotype VI. Please check.	Throughout
20	Impact factors	The original says “as per 2021 update” but newer entries show 5-year figures. Shown as given.	Publications
21	Supervised students	The CV says 2 BS-MS students were guided, but they are not in the supervision table, so they are not in the Alumni list. Send names if you want them added.	Alumni